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EXP 14: PACKET SNIFFING

AIM:

To implement packet sniffing using RAW sockets.

CODE:

```
from scapy.all import sniff
from scapy.layers.inet import IP, TCP, UDP, ICMP
def packet_callback(packet):
    if IP in packet:
        ip_layer = packet[IP]
        protocol = ip_layer.proto
        src_ip = ip_layer.src
        dst_ip = ip_layer.dst
        protocol_name = " "
        if protocol == 1:
            protocol_name = "ICMP"
        elif protocol == 6:
            protocol_name = "TCP"
        elif protocol == 17:
            protocol_name = "UDP"
        else:
            protocol_name = "Unknown Protocol"
        print(f"Protocol : {protocol_name}")
        print(f"Source IP : {src_ip}")
        print(f"Destination IP: {dst_ip}")
        print("-" * 50)
```

```
sniff(
    iface = "wifi",
    prn = packet_callback,
    filter = "ip",
    store = 0
)
```

OUTPUT:

```
python practical14_sniff.py

PROTOCOL : UDP
Source IP : 104.18.41.41
Destination IP : 10.184.154.113

Protocol : UDP
Source IP : 10.14.154.113
Destination IP : 10.184.154.24

Protocol : UDP
Source IP : 10.184.154.113
Destination IP : 10.184.154.24

Protocol : TCP
Source IP : 10.184.154.113
Destination IP : 104.46.162.231.
```

RESULT:

Thus, packet sniffing using RAW sockets is implemented successfully.